

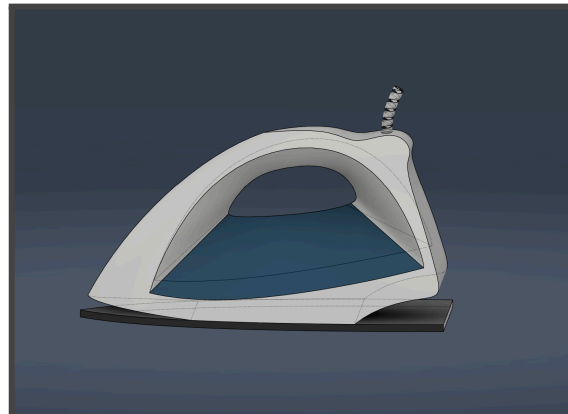
3D Printing and Modeling

8th lab: Surface Modelling

0. Resources

1. [Surface modelling in fusion 360 - AirPods](#) (9 min)
2. [Fusion 360 Surface Modeling Tutorial - Core Concepts Explained](#) (16 min)
3. [Surface Modeling with Fusion 360](#) (19 min)

1. The Iron



The purpose of this laboratory is for you to understand working with surfaces, which will enable you to create more complex shapes than before. To do this we will attempt to replicate the Iron from the image above in the limited time we have.

1.1. Surface Modeling

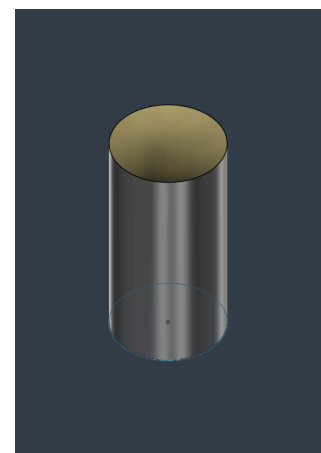
Unlike solid modelling, where every body has volume and thickness, surface modelling works with thin, paper-like geometry (think of it as sculpting with sheets of paper instead of blocks of clay). This makes it powerful for designing complex, more organic shapes (like car panels or bottle contours) where solid tools would struggle to give you the curvature control you need.

The main tools used for creating surfaces are **Extrude & Loft**. You should already be familiar with the first one from the *Solid panel*. For working with surfaces we will need to switch to the *Surfaces panel*. You will have done this correctly if most of the tools in your toolbar are now orange.

Let's try creating a sketch and extruding to see what happens: In the image on the right I extruded from a circle.

You should see that the body has no interior. As mentioned, surfaces have no thickness, so the extrusion is like an infinitely thin sheet of paper wrapped into a cylinder.

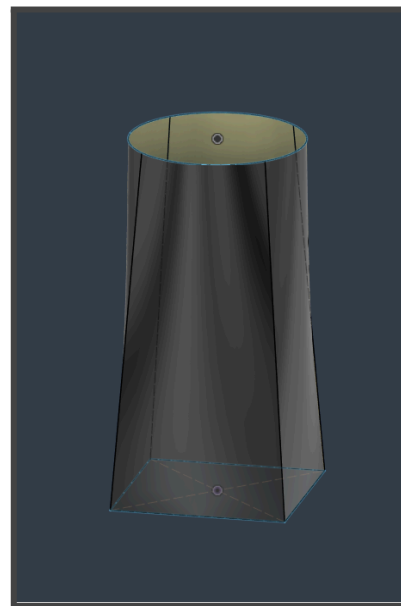
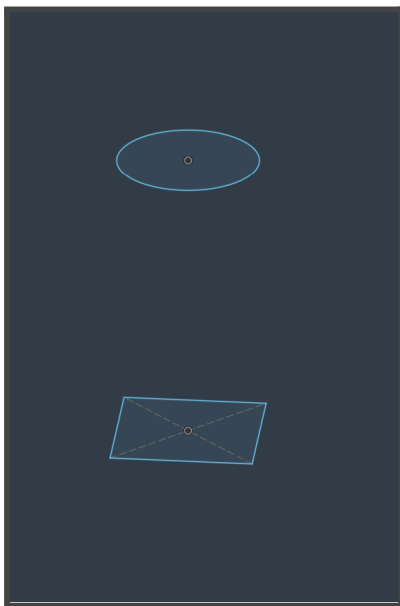
You should also see that it is colored differently than usual. **The gray part represents the exterior of the body, while the yellow will act as the interior.** This will become important when converting between a surface and a solid. You inverse them at any time using **Modify > Reverse Normal**.



1.2. Loft

Loft is today's new addition to your arsenal. It creates a smooth transition between two profiles placed on different planes. For example, you could draw a square on one plane and a circle on another plane, and **Loft** will generate the body that morphs between them. You can further control the shape by adding **guide rails**, which are lines that connect the profiles and constrain how the transition flows between them, similar to the guiding path for **Sweep**.

I strongly suggest you also try to replicate this example and play around with it to get a better understanding of how loft works.

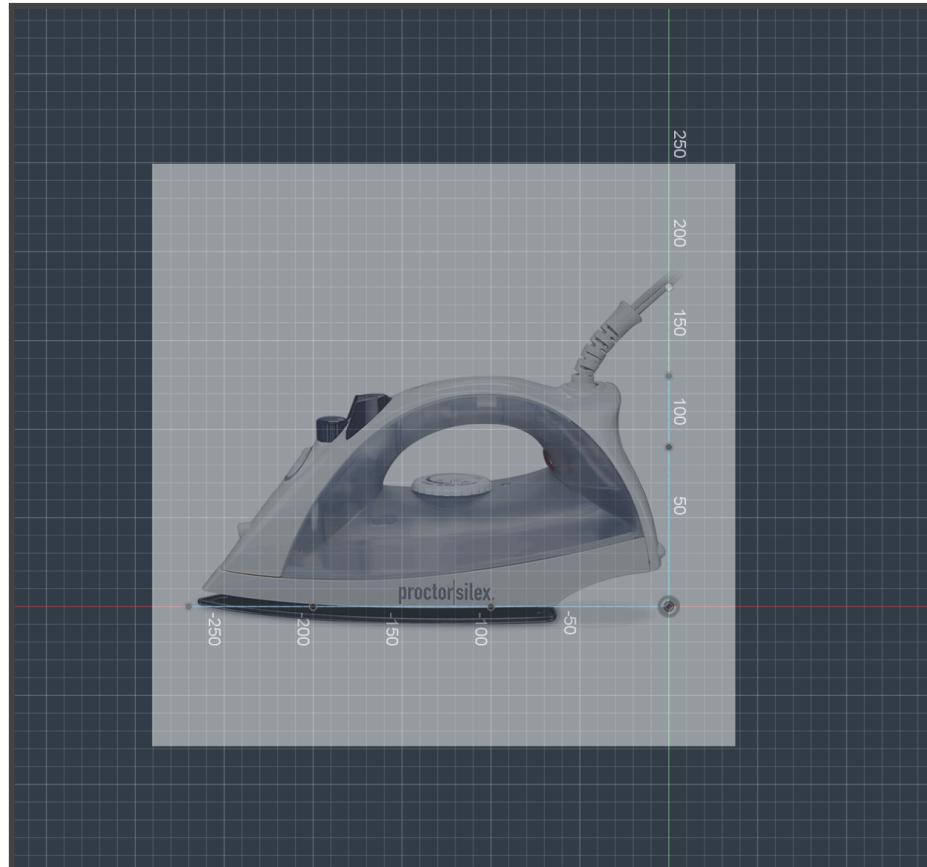


Go to **Create > Loft** and select the shapes you want to connect.

You can also add a thickness to any surface by using **Thicken** (found under **Create**). As the name suggests, it gives the shape a proper volume, turning it into a solid body.

1.3. Inserting a canvas

This time, we will start our design from the picture we wish to replicate. You can download it from teams. We will place it in our design by going to **Insert > Canvas** and selecting the file from our computers, along with the plane on which we want to position it.



I suggest placing in a similar manner, so that the bottom white part sits on one axis, and the tail on another.

The next step will be **scaling** the image to a correct size. Right click the canvas in the browser and select **Calibrate**. Calibrate lets you define a known real-world measurement between two points on an imported canvas image, so Fusion 360 scales it to match the actual dimensions of your object.

For our example, the distance between these two highlighted points should be exactly **20 cm** (200 mm).



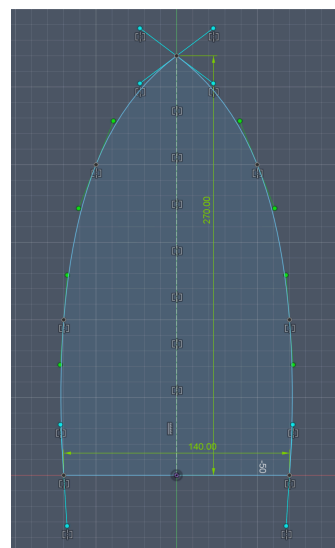
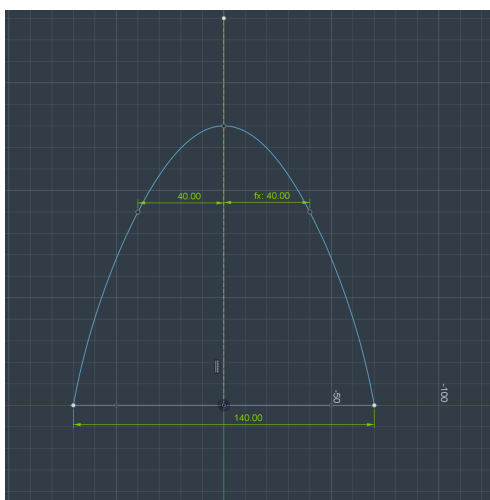
1.4. Sketching

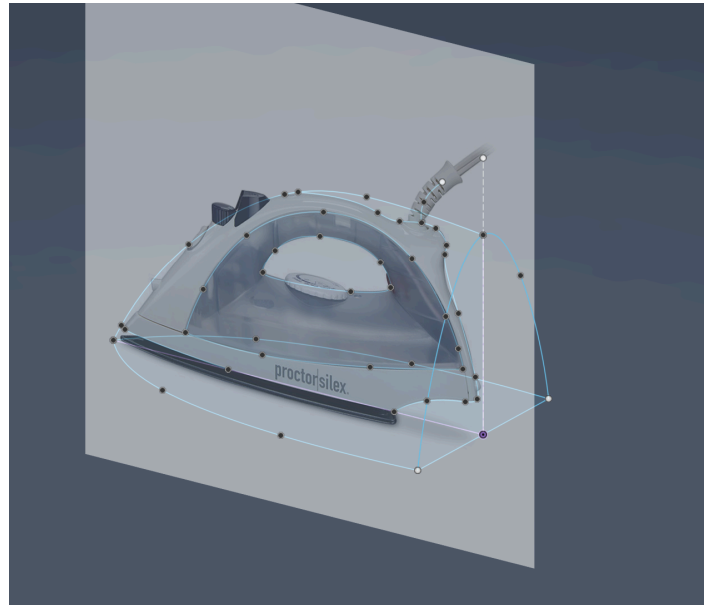


As always, our designs need to start with a sketch. Create a general by following the canvas with the **Fit Point Spline** tool. Just like the image shows, you should keep the exterior outline minimal and connected to the axes. Created another one for the water tank and a small one for the hole in the middle, just like the image shows. They do not need to follow the image perfectly as we can later adjust those sections.

Make sure the exterior outline is connected to the axes. You will run into problems otherwise.

You will also need two other outlines for the back and bottom of the clothes iron. The exact dimensions are not important, but the general shape is.

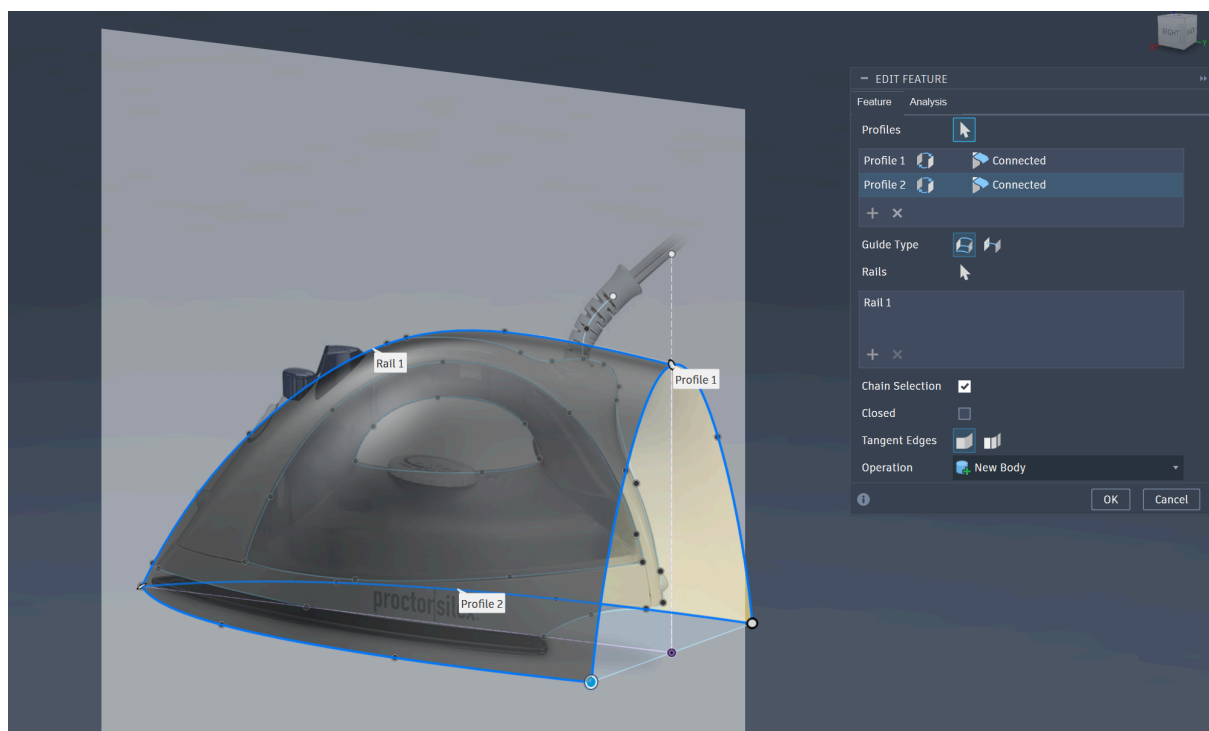




Finished design

1.4. Lofting

The next step is to create the general shape, which we will later adjust. Let's see how loft can help us in this case:

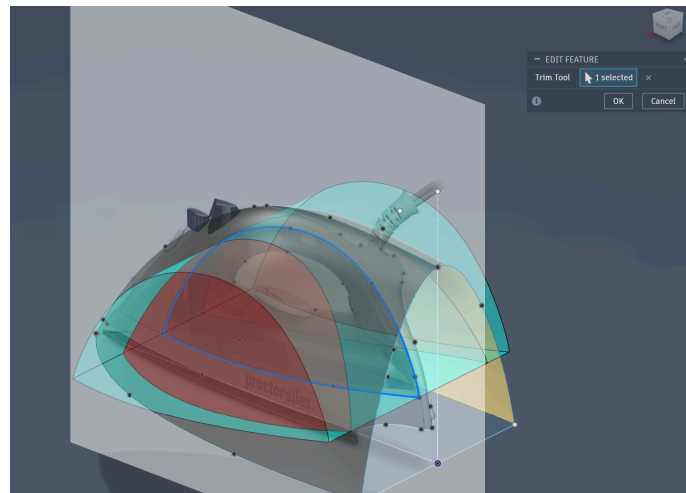


We will create a loft that connects two of our profiles along a given path (called a **Rail**). To do this, we will start by selecting the back and bottom sketches as *Profiles* and the topmost line path in our side view as a rail.

At this point you will encounter an error in the following situations:

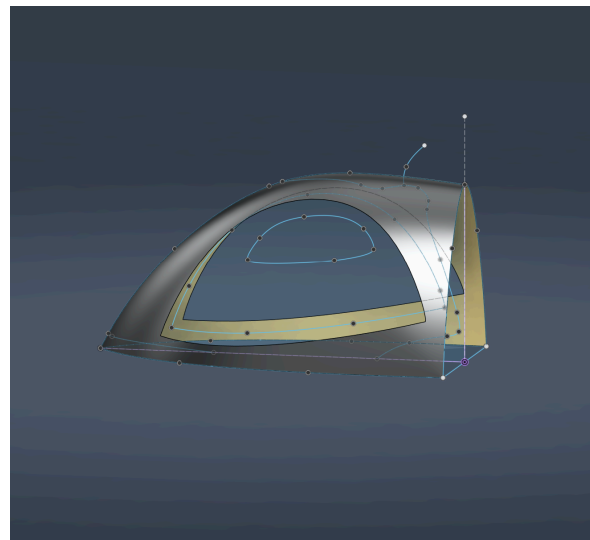
- The rail does not connect to both profiles => adjust the side sketch until it does;
- Geometry can not be created => one of the sketches has a weird shape and Fusion doesn't see a way to connect the two. Adjust the sketches according to the images above.
- Geometry can not be created => can also mean that one of your paths is broken, as in it is composed of multiple lines that are not connected. Fix that.

1.5. Trim



Trim (found under **Modify > Trim**) works similarly to Extrude of type cut. You can select 2 surfaces (or a path and a surface) to eliminate the areas between them. In our case, we want something that looks similar to the image on the right.

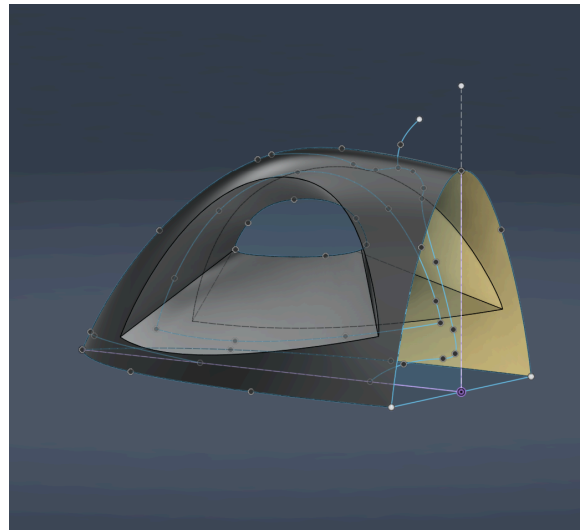
We will achieve this by using the middle outline in our sketch as a cutting plane. Select **Trim** and then the middle path in our side sketch. You should see how it extrudes and cuts our surfaces. The next step is to select the area inside the cutting surface (seen in red in the image above) on both sides.



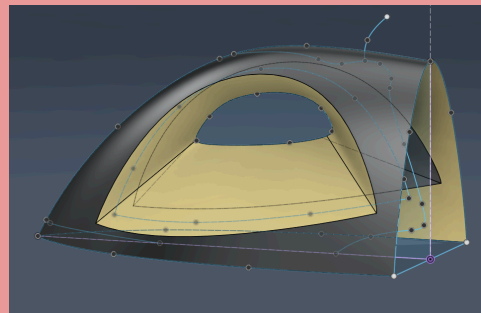
When using trim, you only need to make two selections. The first one is the path or surface that will act as the cutting path, and the second is the surface you wish to remove.

1.6. The handle

Next up is creating the handle. Just like before, we will use **Loft** to stitch together the two planes, like the image below. Click **Loft** and select the outside perimeters of the new surface along with the hole for the handle. We will not be using any guide rails here, but you can add one if you wish.



If Fusion doesn't correctly recognise the inside and outside surfaces, as shown on the right, you can correct this by using **Modify > Reverse Normal** as previously mentioned.



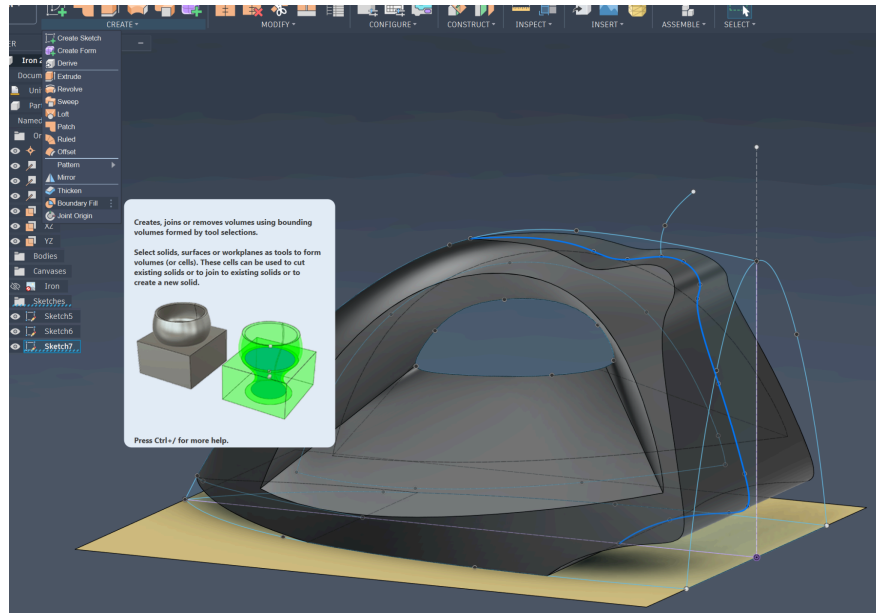
1.7. Finishing the design

If you haven't already, you should now complete the side sketch by adding a small lift at the front and a shape for the back. This will be used to close out our shape and turn it into a body.

Extrude the newly created paths, including the very bottom, until the shape is similar to the one shown below. All that is left is to remove the extra material using **Trim**, and correct any inconsistencies with **Reverse Normal**.



If you followed all the steps, you should have a shape that is theoretically watertight. In that case, one last tool allows us to turn our many surface bodies into one solid body, just like we are used to. Go to **Create > Boundary Fill** and select all surfaces. This tool will only work if your component has no missing surfaces and is completely closed, so be sure to select all of them.



Congratulations!

